

CS 577 — Deep Learning — Homework 2

Read these instructions carefully:

- In the $\text{\LaTeX}$ source code, type your answer in between “`%%% BEGIN ANSWER`” and “`%%% END ANSWER`”. For advanced $\text{\LaTeX}$ users, you can use your custom macros if you wish by placing them between “`%%% BEGIN MACROS`” and “`%%% END MACROS`” in the header. Do not modify anything else.
- Turn in both your `.tex` file and the generated `.pdf` file.

1 Conditional Probability

[3] point(s) — part a: This exercise is designed to help you develop intuition for the concept of conditional probability by “creating one from scratch”.

In mathematics, a *function* takes an input, let’s call it x , to exactly one output, let’s call it y . For example, if $w = 1/2$ and $b = 1/2$, then the function

$$y = \underbrace{f(x)}_{\text{(name of the function)}} = \underbrace{wx + b}_{\text{(what the function actually is)}} \quad (1)$$

assigns every input x to exactly one real number, namely $wx + b$. In this case, we say the dependency of y on x , i.e., the input-output relationship, is *deterministic*. Note in naming the function, we could have used g or h instead of f . There is nothing special about the letter “ f ” other than tradition (and also that the word “function” starts with that letter).

In the real world, it is often the case that the dependency of y on x has a degree of uncertainty. This notion is what *conditional probability* aims to capture:

$$\underbrace{p(y \mid x)}_{\text{(name of the conditional probability)}} = \underbrace{\hspace{2cm}}_{\text{(what the conditional probability actually is)}} \quad \text{(Intentionally blank)}$$

Just like for functions, there is nothing special about the letter p ; we could’ve wrote $q(y \mid \mathbf{x})$ or $r(y \mid \mathbf{x})$. Now, why did we leave the right-hand-side blank? We’ve learned how to “create” functions since grade school mathematics. But most of us never learned the notation for how to do the same for conditional probability. Let’s do that here.

Let $n > 0$ be an integer. Consider

$$y = wx + b + \epsilon$$

where w and b are non-random numbers and x is the input, but ϵ is random and defined via the following algorithm:

1. Initialize $\epsilon \leftarrow 0$
2. For $t = 1, \dots, n$, do:
 1. Flip a fair coin
 2. If coin comes up head, then $\epsilon \leftarrow \epsilon + 1$
 3. Else $\epsilon \leftarrow \epsilon - 1$
3. Return $\epsilon/\sqrt{n}$.

The above procedure defines a dependency of y on x that is random, and hence the input-output relationship is governed by a conditional probability. Let's call the conditional probability

$$\mathcal{S}(y; wx + b, n)$$

(the curly “S” is $\mathcal{S}$). When $n = 2$, we have

$$p(y | x) = \mathcal{S}(y; wx + b, 2) = \begin{cases} 1/4 & : \text{if } y = wx + b + \frac{2}{\sqrt{2}}, \\ 1/2 & : \text{if } y = wx + b \\ 1/4 & : \text{if } y = wx + b - \frac{2}{\sqrt{2}}, \\ 0 & : \text{otherwise.} \end{cases}$$

To see there, first note that there are 4 possible outcomes for the coin flip. There is only one outcome in which both coins come up heads, then $y = wx + b + \frac{2}{\sqrt{2}}$. Thus, the probability is $1/4$. The same logic applies to the $y = wx + b - \frac{2}{\sqrt{2}}$ case. Now, there are two outcomes that results in $y = wx + b$: one coin comes up head while the other comes up tail. Hence the probability of this case is $2/4 = 1/2$. It is impossible for y to equal to any other values.

The exercise: calculate $p(y | x) = \mathcal{S}(y; wx + b, 3)$. *Hint: introduce a notation for denoting outcomes of the coin flips.*

Answer:

1: Each flip can result in either a head (H) or a tail (T). For $n = 3$, there are $2^3 = 8$ possible outcomes:

HHH, HHT, HTH, HTT, THH, THT, TTH, TTT.

2: The steps for determining ϵ is: - Start with $\epsilon = 0$, - Add $+1$ for each head, - Subtract -1 for each tail, - Return $\epsilon/\sqrt{3}$.

Let's find ϵ for each outcome:

$$\begin{aligned} \text{HHH} : \epsilon = 3 &\Rightarrow \frac{3}{\sqrt{3}} = \sqrt{3}, \\ \text{HHT} : \epsilon = 1 &\Rightarrow \frac{1}{\sqrt{3}} = \frac{1}{\sqrt{3}}, \\ \text{HTH} : \epsilon = 1 &\Rightarrow \frac{1}{\sqrt{3}} = \frac{1}{\sqrt{3}}, \\ \text{HTT} : \epsilon = -1 &\Rightarrow \frac{-1}{\sqrt{3}} = -\frac{1}{\sqrt{3}}, \\ \text{THH} : \epsilon = 1 &\Rightarrow \frac{1}{\sqrt{3}} = \frac{1}{\sqrt{3}}, \\ \text{THT} : \epsilon = -1 &\Rightarrow \frac{-1}{\sqrt{3}} = -\frac{1}{\sqrt{3}}, \\ \text{TTH} : \epsilon = -1 &\Rightarrow \frac{-1}{\sqrt{3}} = -\frac{1}{\sqrt{3}}, \\ \text{TTT} : \epsilon = -3 &\Rightarrow \frac{-3}{\sqrt{3}} = -\sqrt{3}. \end{aligned}$$

Using the formula $y = wx + b + \epsilon$, we get the possible values of y based on ϵ :

$$\begin{aligned} y &= wx + b + \sqrt{3}, \\ y &= wx + b + \frac{1}{\sqrt{3}}, \\ y &= wx + b - \frac{1}{\sqrt{3}}, \\ y &= wx + b - \sqrt{3}. \end{aligned}$$

4: There are 8 possible outcomes from the coin flips, and the probabilities are:

$$\begin{aligned} \mathbb{P}(y = wx + b + \sqrt{3}) &= \frac{1}{8} \quad (\text{HHH}), \\ \mathbb{P}(y = wx + b + \frac{1}{\sqrt{3}}) &= \frac{3}{8} \quad (\text{HHT, HTH, THH}), \\ \mathbb{P}(y = wx + b - \frac{1}{\sqrt{3}}) &= \frac{3}{8} \quad (\text{HTT, THT, TTH}), \\ \mathbb{P}(y = wx + b - \sqrt{3}) &= \frac{1}{8} \quad (\text{TTT}). \end{aligned}$$

Final Answer: Thus, the conditional probability $p(y|x) = \mathcal{S}(y; wx + b, 3)$ is:

$$p(y|x) = \mathcal{S}(y; wx + b, 3) = \begin{cases} \frac{1}{8}, & \text{if } y = wx + b + \sqrt{3}, \\ \frac{3}{8}, & \text{if } y = wx + b + \frac{1}{\sqrt{3}}, \\ \frac{3}{8}, & \text{if } y = wx + b - \frac{1}{\sqrt{3}}, \\ \frac{1}{8}, & \text{if } y = wx + b - \sqrt{3}, \\ 0, & \text{otherwise.} \end{cases}$$

[2] point(s) — part b: [This is a programming exercise. See hw2.ipynb](#)

[2] point(s) — part c: [This is a programming exercise. See hw2.ipynb](#)

2 The Gaussian distribution

[1] point(s) — part a: Consider linear regression when the data live in the 2-dimensional space. That is, the data vectors $\mathbf{x}^{(i)} \in \mathbb{R}^2$ are 2-dimensional vectors, and the labels $y \in \mathbb{R}$ are numbers.

In lecture 2, we saw that for linear regression, the relationship between the sample (input) $\mathbf{x}$ and the label (output) y is governed by

$$p(y \mid \mathbf{x}) = \mathcal{N}(y; \mathbf{w}^\top \mathbf{x} + b, \sigma^2) \quad (2)$$

where $\mathcal{N}(\cdot; \mu, \sigma^2)$ is the probability density function for a Gaussian random variable with mean μ and variance σ^2 . The formula for $\mathcal{N}(\cdot; \mu, \sigma^2)$ is given by

$$\mathcal{N}(z; \mu, \sigma^2) := \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2} \frac{(z - \mu)^2}{\sigma^2}\right).$$

Let $\sigma = 1/\sqrt{2}$, $\mathbf{w} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and $b = 3$. Compute $p(y \mid \mathbf{x})$ when $\mathbf{x} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$. The only unknown variable in your solution should be y . Type your answer in a single line of L^AT_EX code using fewer than 50 characters. You don't need to show your work.

Answer:

$$p(y \mid x) = \frac{1}{\sqrt{\pi}} \exp(-(y - 1)^2).$$

[1] point(s) — part b: Plug in $y = 1$ and $y = -1/2$ into your answer above. Compute the numerical value of $\frac{p(1|\mathbf{x})}{p(-1/2|\mathbf{x})}$ and write the result below rounded to three decimal places (i.e., like 1.234)

Answer:

$$\frac{p(1|x)}{p(-\frac{1}{2}|x)} \approx 9.559$$

[1] point(s) — part c: [This is a programming exercise. See hw2.ipynb](#)

[2] point(s) — part d: “ $p(y \mid \mathbf{x}) = \mathcal{N}(y; \mathbf{w}^\top \mathbf{x} + b, \sigma^2)$ ” is what is known as a probability *density*: $p(y \mid \mathbf{x})$ is not the probability of observing exactly y , but rather the probability of observing a value *near* y . The number in the output cell of part c should be similar to the number in your answer in part b. Explain this why these two numbers are similar.

Answer:

The probability density function (PDF) $p(y \mid x)$ is like a guide that tells us how likely it is to see a value close to y . In part c, when we look at the ratio $n1/n2$, we're figuring out how often we see values of y within certain ranges. In part b, the ratio $p(1 \mid x)/p(-1/2 \mid x)$ is about comparing how likely different specific values of y are.

Both of these ratios are similar because they both give us an idea of how likely it is to see certain values of y given x . The PDF gives us a theoretical view of this likelihood, while the empirical frequency shows us the actual data we've observed. In essence, both are ways to understand the same thing: how likely different values of y are when x is known.

3 A nonlinear regression problem

[3] **point(s)** — **part a:** Consider samples $x \in \mathbb{R}$ and labels $y \in \mathbb{R}$. Let $\boldsymbol{\theta} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$ and let $f(x; \boldsymbol{\theta}) = \sin(w_1 x) + \cos(w_2 x)$. Calculate the gradient $\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta})$ of the function

$$J(\boldsymbol{\theta}) = J\left(\begin{bmatrix} w_1 \\ w_2 \end{bmatrix}\right)^2 = \frac{1}{N} \sum_{i=1}^N (y^{(i)} - f(\mathbf{x}^{(i)}; \boldsymbol{\theta}))^2.$$

Answer:

Lets compute the gradient $\nabla_{\theta}J(\theta)$ of the function

$$J(\theta) = \frac{1}{N} \sum_{i=1}^N \left(y^{(i)} - f(x^{(i)}; \theta) \right)^2,$$

where

$$f(x; \theta) = \sin(w_1 x) + \cos(w_2 x),$$

and $\theta = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$.

1. Partial Derivatives of $f(x; \theta)$ with respect to w_1 and w_2 :

$$\frac{\partial f(x; \theta)}{\partial w_1} = \frac{\partial}{\partial w_1} (\sin(w_1 x) + \cos(w_2 x)) = x \cos(w_1 x).$$

$$\frac{\partial f(x; \theta)}{\partial w_2} = \frac{\partial}{\partial w_2} (\sin(w_1 x) + \cos(w_2 x)) = -x \sin(w_2 x).$$

2. Applying the chain rule to get the gradient ($\nabla_{\theta}J(\theta)$) :

$$\nabla_{\theta}J(\theta) = \begin{bmatrix} \frac{\partial J(\theta)}{\partial w_1} \\ \frac{\partial J(\theta)}{\partial w_2} \end{bmatrix}.$$

For the loss $J(\theta)$, we have:

$$\frac{\partial J(\theta)}{\partial w_1} = -\frac{2}{N} \sum_{i=1}^N \left(y^{(i)} - f(x^{(i)}; \theta) \right) \frac{\partial f(x^{(i)}; \theta)}{\partial w_1},$$

$$\frac{\partial J(\theta)}{\partial w_2} = -\frac{2}{N} \sum_{i=1}^N \left(y^{(i)} - f(x^{(i)}; \theta) \right) \frac{\partial f(x^{(i)}; \theta)}{\partial w_2}.$$

On Substituting the partial derivatives:

$$\frac{\partial J(\theta)}{\partial w_1} = -\frac{2}{N} \sum_{i=1}^N \left(y^{(i)} - f(x^{(i)}; \theta) \right) x^{(i)} \cos(w_1 x^{(i)}),$$

$$\frac{\partial J(\theta)}{\partial w_2} = -\frac{2}{N} \sum_{i=1}^N \left(y^{(i)} - f(x^{(i)}; \theta) \right) (-x^{(i)} \sin(w_2 x^{(i)})).$$

This implies to:

$$\frac{\partial J(\theta)}{\partial w_1} = -\frac{2}{N} \sum_{i=1}^N \left(y^{(i)} - f(x^{(i)}; \theta) \right) x^{(i)} \cos(w_1 x^{(i)}),$$

$$\frac{\partial J(\theta)}{\partial w_2} = \frac{2}{N} \sum_{i=1}^N \left(y^{(i)} - f(x^{(i)}; \theta) \right) x^{(i)} \sin(w_2 x^{(i)}).$$

3. The gradient $\nabla_{\theta}J(\theta)$ is:

$$\nabla_{\theta}J(\theta) = \begin{bmatrix} -\frac{2}{N} \sum_{i=1}^N \left(y^{(i)} - f(x^{(i)}; \theta) \right) x^{(i)} \cos(w_1 x^{(i)}) \\ \frac{2}{N} \sum_{i=1}^N \left(y^{(i)} - f(x^{(i)}; \theta) \right) x^{(i)} \sin(w_2 x^{(i)}) \end{bmatrix}.$$

[1] **point(s)** — **part b:** [This is a programming exercise. See hw2.ipynb](#)

[4] **point(s)** — **part c:** Run gradient descent for 100 steps with constant step size = $1/4$. Compare two different initialization: initialization “A”

$$w_1 = -2.5, \quad w_2 = 3.56$$

and initialization “B”

$$w_1 = -2.5, \quad w_2 = 3.57.$$

Answer the following questions:

1. Which initialization led to better performance in terms of the training error?
2. Is one of the converged final parameter (nearly) optimal? How do you know?

For this part, only your response below will be graded. We will not take into consideration of the contents in the python notebook.

Answer:

1. **Which initialization led to better performance in terms of the training error?**

Initialization B performed better, achieving a much lower final MSE of 0.0075 compared to Initialization A's final MSE of 1.0072. This indicates a more effective convergence.

2. **Is one of the converged final parameters (nearly) optimal? How do you know?**

Yes, the final parameters from Initialization B appear nearly optimal. The MSE is very close to zero, suggesting that the model parameters have converged to a point minimizing the training error effectively.

[10 bonus] **point(s)** — **part d:** Explain in as much quantitative detail as possible why these two initialization led to different results. If you wish to use figures, you may use

`\includegraphics[width=0.5\textwidth]{myfigure.png}`

Do not submit the figures separately.

For this part, only your response below will be graded. We will not take into consideration of the contents in the python notebook.

Answer:

Quantitative Explanation for Different Results:

1. Initialization A converged to a final MSE of 1.0072, whereas Initialization B achieved a much lower final MSE of 0.0075. This indicates that Initialization B found a more optimal set of parameters.
2. Between iterations 10 and 20, Initialization B showed a sharp decrease in MSE from 1.1119 to 0.1855, while Initialization A's MSE only decreased marginally from 1.1419 to 1.0073 in the same range.
3. The steeper descent in Initialization B suggests it started closer to a region with a more favorable gradient, leading to faster convergence. Initialization A plateaued earlier, showing less improvement over iterations.
4. By iteration 25, Initialization B had already reached an MSE of 0.0075, while Initialization A was still at 1.0072, indicating that B found a better local minimum much faster.

Running Gradient Descent with Initialization A ($w_1 = -2.5$, $b_1 = 3.56$)

```
Iteration 0: MSE = 2.3072, w = -2.4960, b = 3.6051
Iteration 5: MSE = 1.4088, w = -2.5688, b = 4.4586
Iteration 10: MSE = 1.1419, w = -2.5314, b = 4.7928
Iteration 15: MSE = 1.0431, w = -2.2176, b = 4.7821
Iteration 20: MSE = 1.0073, w = -2.0244, b = 4.7761
Iteration 25: MSE = 1.0072, w = -2.0101, b = 4.7758
Iteration 30: MSE = 1.0072, w = -2.0088, b = 4.7758
Iteration 35: MSE = 1.0072, w = -2.0087, b = 4.7758
Iteration 40: MSE = 1.0072, w = -2.0087, b = 4.7758
Iteration 45: MSE = 1.0072, w = -2.0087, b = 4.7758
Iteration 50: MSE = 1.0072, w = -2.0087, b = 4.7758
Iteration 55: MSE = 1.0072, w = -2.0087, b = 4.7758
Iteration 60: MSE = 1.0072, w = -2.0087, b = 4.7758
Iteration 65: MSE = 1.0072, w = -2.0087, b = 4.7758
Iteration 70: MSE = 1.0072, w = -2.0087, b = 4.7758
Iteration 75: MSE = 1.0072, w = -2.0087, b = 4.7758
Iteration 80: MSE = 1.0072, w = -2.0087, b = 4.7758
Iteration 85: MSE = 1.0072, w = -2.0087, b = 4.7758
Iteration 90: MSE = 1.0072, w = -2.0087, b = 4.7758
Iteration 95: MSE = 1.0072, w = -2.0087, b = 4.7758
Final MSE = 1.0072
Final weights (A): w = -2.0087, b = 4.7758
```

Running Gradient Descent with Initialization B ($w_2 = -2.5$, $b_2 = 3.57$)

```
Iteration 0: MSE = 2.3032, w = -2.4972, b = 3.6197
Iteration 5: MSE = 1.2879, w = -2.5951, b = 4.5551
Iteration 10: MSE = 1.1427, w = -2.6540, b = 4.7894
Iteration 15: MSE = 1.1119, w = -2.8321, b = 4.7711
Iteration 20: MSE = 0.1855, w = -3.7100, b = 4.7512
Iteration 25: MSE = 0.0075, w = -4.0002, b = 4.8024
Iteration 30: MSE = 0.0075, w = -4.0002, b = 4.8024
Iteration 35: MSE = 0.0075, w = -4.0002, b = 4.8024
Iteration 40: MSE = 0.0075, w = -4.0002, b = 4.8024
Iteration 45: MSE = 0.0075, w = -4.0002, b = 4.8024
Iteration 50: MSE = 0.0075, w = -4.0002, b = 4.8024
Iteration 55: MSE = 0.0075, w = -4.0002, b = 4.8024
Iteration 60: MSE = 0.0075, w = -4.0002, b = 4.8024
Iteration 65: MSE = 0.0075, w = -4.0002, b = 4.8024
Iteration 70: MSE = 0.0075, w = -4.0002, b = 4.8024
Iteration 75: MSE = 0.0075, w = -4.0002, b = 4.8024
Iteration 80: MSE = 0.0075, w = -4.0002, b = 4.8024
Iteration 85: MSE = 0.0075, w = -4.0002, b = 4.8024
```